Atelier Python GUI & MVC

Introduction à Tkinter, PySide6 et l'architecture MVC

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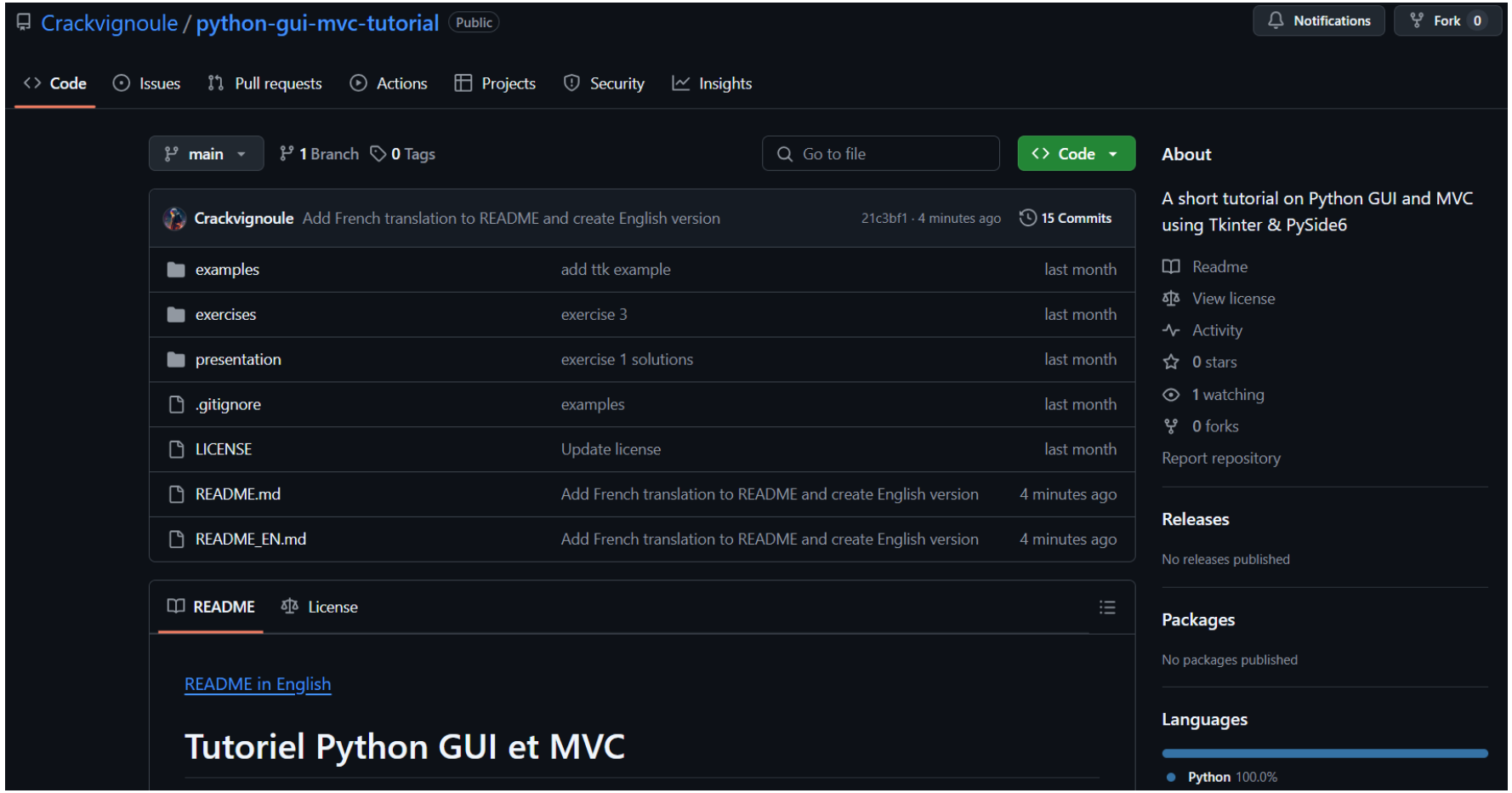
Plan

1. **Initiation aux interfaces graphiques en Python**
   a. **Bibliothèques graphiques Tkinter & PySide6**
   b. **Étapes de création d'une GUI**
   c. **Widgets et Layouts**
   d. **Exemples**
   e. **Utilisation QTDesigner (optionnel)**
   f. **Exercices**
2. **Introduction MVC**
   a. **Explication du concept**
   b. **Exemples**

Lien du repo GitHub

bit.ly/4f3jjm7



1. Initiation aux interfaces graphiques en Python

Initiation aux interfaces graphiques en Python

Bibliothèques graphiques Tkinter & PySide6

Bibliothèque native	Interface moderne et responsive
Bibliothèque très légère	Large collection de widgets
Simple à utiliser pour projets légers	Documentation complète
Exemples abondants sur internet	Fonctions supplémentaires (drag’n drop, signaux…)
	Ecosystème Qt, ex: Concepteur Graphique « QTDesigner »

Critère	Tkinter	PySide6
Facilité d’apprentissage	★★★★★	★★★
Puissance et flexibilité	★★★	★★★★★
Apparence native et cohérence multi-plateformes	★★	★★★★★

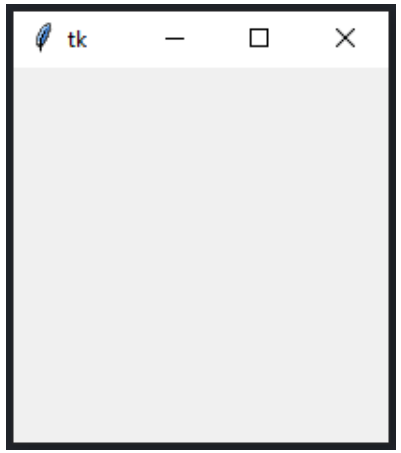
Initiation aux interfaces graphiques en Python

Étapes de création d'une GUI

1. **Importer la bibliothèque**
2. **Créer la fenêtre principale (Main Window)**
3. **Ajouter des widgets (boutons, textes…)**
4. **Entrer dans la boucle d'évènements (Event Loop)**

Tkinter

```python
import tkinter as tk  # Importer tkinter
root = tk.Tk()        # Créer Main Window
root.mainloop()       # Lancer la boucle principale
```



PySide6

```python
from PySide6.QtWidgets import QApplication, QWidget
app = QApplication([])  # Créer une application
window = QWidget()      # Créer une fenêtre
window.show()           # Afficher la fenêtre
app.exec()             # Lancer la boucle principale
```

Initiation aux interfaces graphiques en Python

Étapes de création d'une GUI

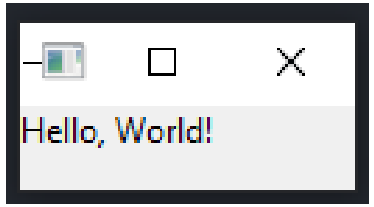
1. Importer la bibliothèque
2. Créer la fenêtre principale (Main Window)
3. **Ajouter des widgets (boutons, textes…)**
4. Entrer dans la boucle d'évènements (Event Loop)

Tkinter

```python
import tkinter as tk  # Importer tkinter
root = tk.Tk()        # Créer Main Window
label = tk.Label(root, text="Hello, World!")  # Créer un label
label.pack()                                  # Ajouter le label
root.mainloop()       # Lancer la boucle principale
```

PySide6

```python
from PySide6.QtWidgets import QApplication, QWidget, QLabel
app = QApplication([])  # Créer une application
window = QWidget()      # Créer une fenêtre
label = QLabel("Hello, World!", parent=window)  # Créer un label
window.show()           # Afficher la fenêtre
app.exec()             # Lancer la boucle principale
```



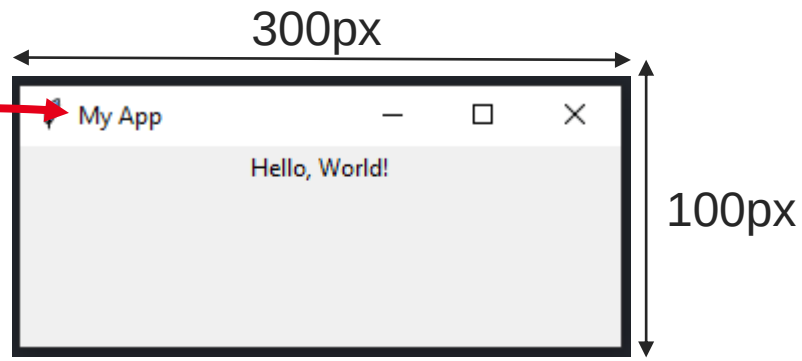
Initiation aux interfaces graphiques en Python

Étapes de création d'une GUI

(optionnel) Changer le titre et la taille de la fenêtre

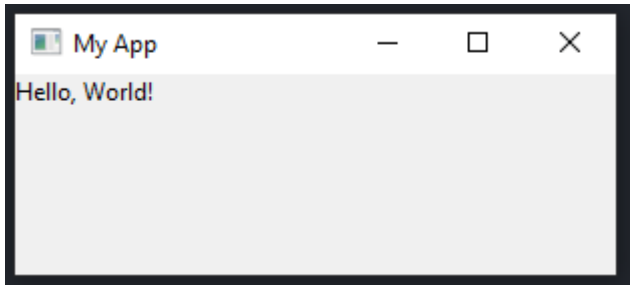
Tkinter

```python
import tkinter as tk  # Importer tkinter
root = tk.Tk()        # Créer Main Window
root.title("My App")      # Changer le titre de la fenêtre
root.geometry("300x100")  # Définir la taille de la fenêtre
label = tk.Label(root, text="Hello, World!")  # Créer un label
label.pack()                                  # Ajouter le label
root.mainloop()       # Lancer la boucle principale
```



PySide6

```python
from PySide6.QtWidgets import QApplication, QWidget, QLabel
app = QApplication([])  # Créer une application
window = QWidget()      # Créer une fenêtre
window.setWindowTitle("My App")  # Changer le titre de la fenêtre
window.setGeometry(0, 0, 300, 100)  # Définir la taille de la fenêtre
label = QLabel("Hello, World!", parent=window)  # Créer un label
window.show()           # Afficher la fenêtre
app.exec()             # Lancer la boucle principale
```

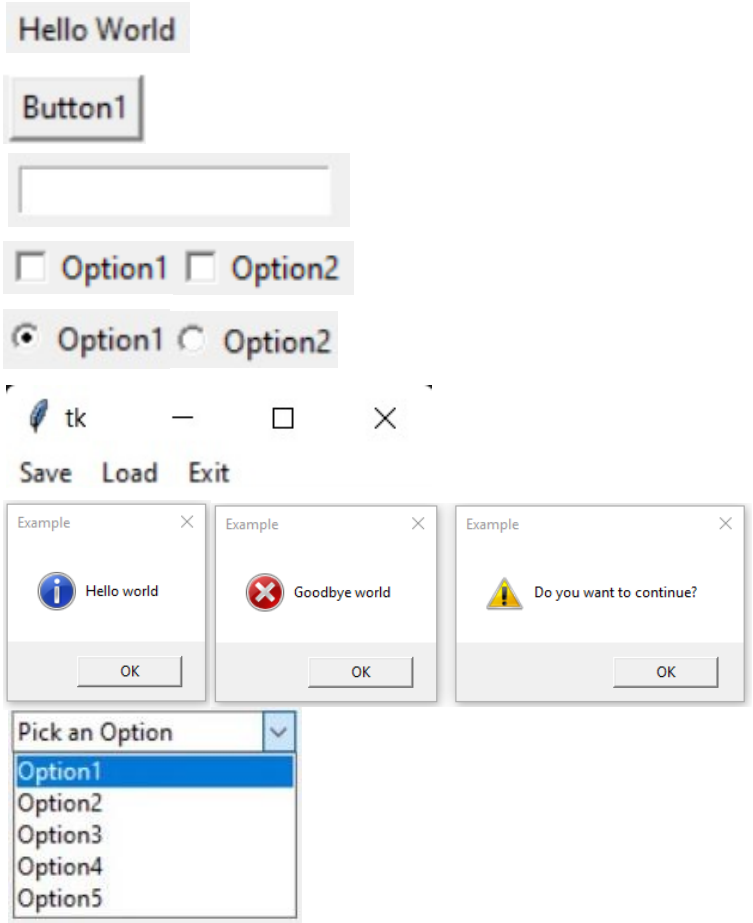


Initiation aux interfaces graphiques en Python

Widgets et Layouts

Une liste non exhaustive de widgets:

Widget	Tkinter	PySide6
Label	tk.Label(text="...")	QLabel("...")
Bouton	tk.Button(text="...")	QPushButton("...")
Entry	tk.Entry	QLineEdit
Checkbox	tk.Checkbutton	QCheckBox
Radiobutton	tk.Radiobutton	QRadioButton
Menu	tk.Menu	QMenuBar
MessageBox	tk.messagebox	QMessageBox
Combobox	from tkinter import ttk ttk.Combobox	QComboBox

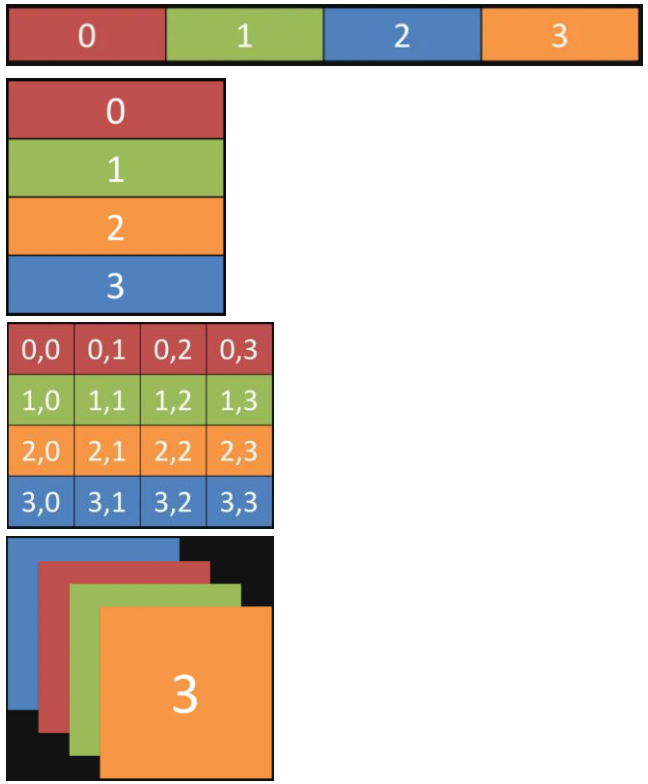


Initiation aux interfaces graphiques en Python

Widgets et Layouts

Layout: Gestionnaire de disposition qui organise les widgets dans une interface utilisateur.

Layout	Tkinter	Description
QHBoxLayout	widget.pack(side="left")	Ligne Horizontale
QVBoxLayout	widget.pack(side="top")	Ligne Verticale
QGridLayout	widget.grid(row=0, column=1)	Grille X,Y
QStackedLayout	Pas d'équivalent direct	Stack empilé Z



source: https://www.pythonguis.com/tutorials/pyside6-layouts/

Initiation aux interfaces graphiques en Python

Exemples

bit.ly/4f3jjm7

GitHub

1. Ouvrir l'invite de commande (cmd) sur Windows:
   - Appuyez sur `Win + R` pour ouvrir la boîte de dialogue Exécuter.
   - Tapez `cmd` et appuyez sur `Entrée`.
2. Cloner le dépôt :

```
git clone https://github.com/Crackvignoule/python-gui-mvc-tutorial.git
cd python-gui-mvc-tutorial
```

3. Créer un environnement virtuel :

```
python -m venv .venv
```

4. Activer l'environnement virtuel :
   - Sur Windows :

```
.\.venv\Scripts\activate
```

   - Sur macOS et Linux :

```
source .venv/bin/activate
```

5. Installer les dépendances :

```
pip install -r requirements.txt
```

Initiation aux interfaces graphiques en Python

Exemples

Utilisation QMainWindow + Layout

PySide6

```
app = QApplication([])
window = QMainWindow()

central_widget = QWidget()
layout = QVBoxLayout()
```

⋮

```
layout.addWidget(email_label)
layout.addWidget(email_input)
layout.addWidget(submit_button)

central_widget.setLayout(layout)
window.setCentralWidget(central_widget)
```

Tkinter

```
label.pack()
entry.pack()
button.pack()
```

```
label.grid(row=0, column=0)
entry.grid(row=0, column=1)
button.grid(row=1, column=0)
```

Initiation aux interfaces graphiques en Python

Exemples

Utilisation Button:

Tkinter

```
tk.Button(root, text="Submit", command=submit)
```

PySide6

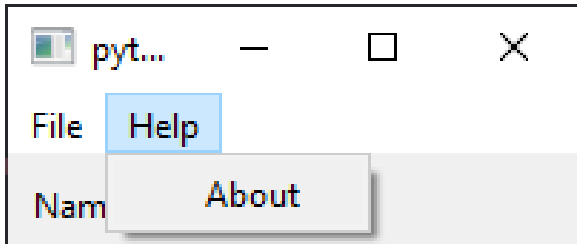
```
submit_button = QPushButton("Submit")
submit_button.clicked.connect(submit)
```

```
def submit():
    print("...")
```

Initiation aux interfaces graphiques en Python

Exemples

Utilisation MenuBar:



Tkinter

```
menu_bar = tk.Menu(root)
root.config(menu=menu_bar)
```

PySide6

```
menu_bar = QMenuBar()
window.setMenuBar(menu_bar)
```

```
help_menu = tk.Menu(menu_bar, tearoff=0)
menu_bar.add_cascade(label="Help", menu=help_menu)
help_menu.add_command(label="About", command=show_about)
```

```
help_menu = menu_bar.addMenu("Help")
about_action = QAction("About", window)
about_action.triggered.connect(show_about)
help_menu.addAction(about_action)
```

Initiation aux interfaces graphiques en Python

Exemples

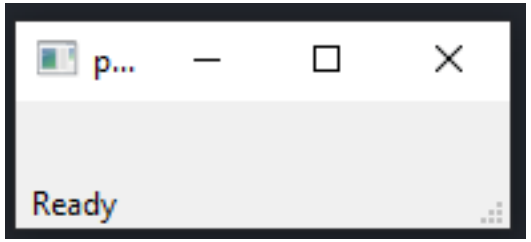
Utilisation StatusBar

Tkinter

```
status_var = tk.StringVar()
status_var.set("Ready")
status_bar = tk.Label(root, textvariable=status_var, bd=1, anchor=tk.W)
status_bar.grid(row=4, columnspan=2, sticky=tk.W+tk.E)
```

PySide6

```
status_bar = QStatusBar()
window.setStatusBar(status_bar)
status_bar.showMessage("Ready")
```

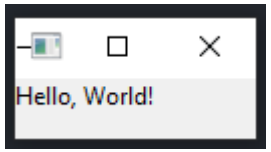


Initiation aux interfaces graphiques en Python

Exemples

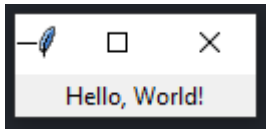
Tkinter

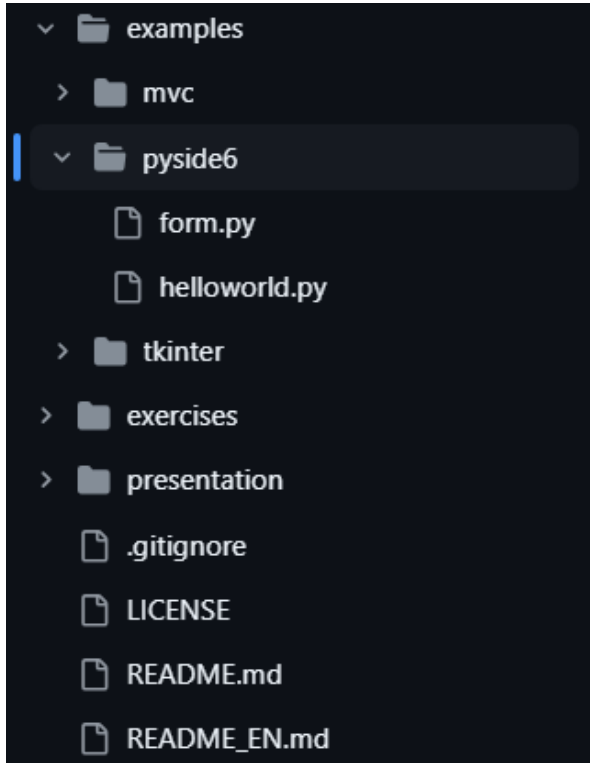
```
python ./examples/tkinter/helloworld.py
```



PySide6

```
python ./examples/pyside6/helloworld.py
```



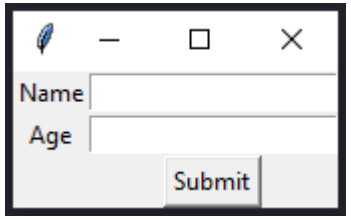


Initiation aux interfaces graphiques en Python

Exemples

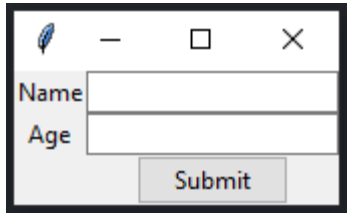
Tkinter

```
python ./examples/tkinter/form.py
```



Tkinter

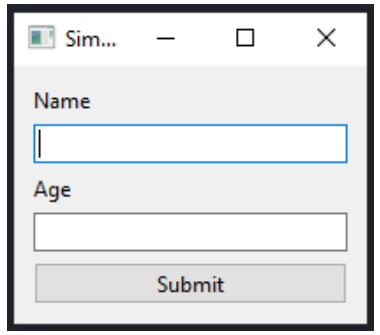
```
python ./examples/tkinter/ttk_form.py
```




```python
import tkinter as tk
from tkinter import ttk        You, 2 months ago • add ttk example

def submit():
    print(f"Name: {name_entry.get()}, Age: {age_entry.get()}")

root = tk.Tk()
root.title("Simple Form with ttk")

ttk.Label(root, text="Name").grid(row=0)
ttk.Label(root, text="Age").grid(row=1)

name_entry = ttk.Entry(root)
age_entry = ttk.Entry(root)

name_entry.grid(row=0, column=1)
age_entry.grid(row=1, column=1)

ttk.Button(root, text="Submit", command=submit).grid(row=2, column=1)

root.mainloop()
```

PySide6

```
python ./examples/pyside6/form.py
```

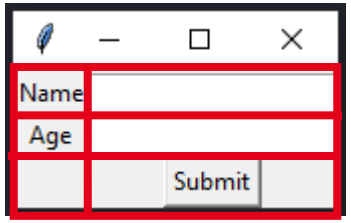


Initiation aux interfaces graphiques en Python

Exemples

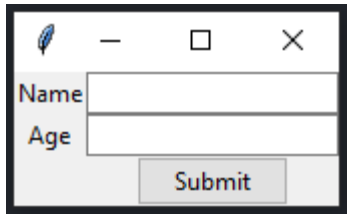
Tkinter

```
python ./examples/tkinter/form.py
```



Tkinter

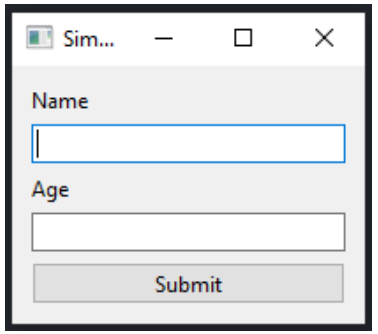
```
python ./examples/tkinter/ttk_form.py
```




```python
import tkinter as tk
from tkinter import ttk        You, 2 months ago • add ttk example

def submit():
    print(f"Name: {name_entry.get()}, Age: {age_entry.get()}")

root = tk.Tk()
root.title("Simple Form with ttk")

ttk.Label(root, text="Name").grid(row=0)
ttk.Label(root, text="Age").grid(row=1)

name_entry = ttk.Entry(root)
age_entry = ttk.Entry(root)

name_entry.grid(row=0, column=1)
age_entry.grid(row=1, column=1)

ttk.Button(root, text="Submit", command=submit).grid(row=2, column=1)

root.mainloop()
```

PySide6

```
python ./examples/pyside6/form.py
```

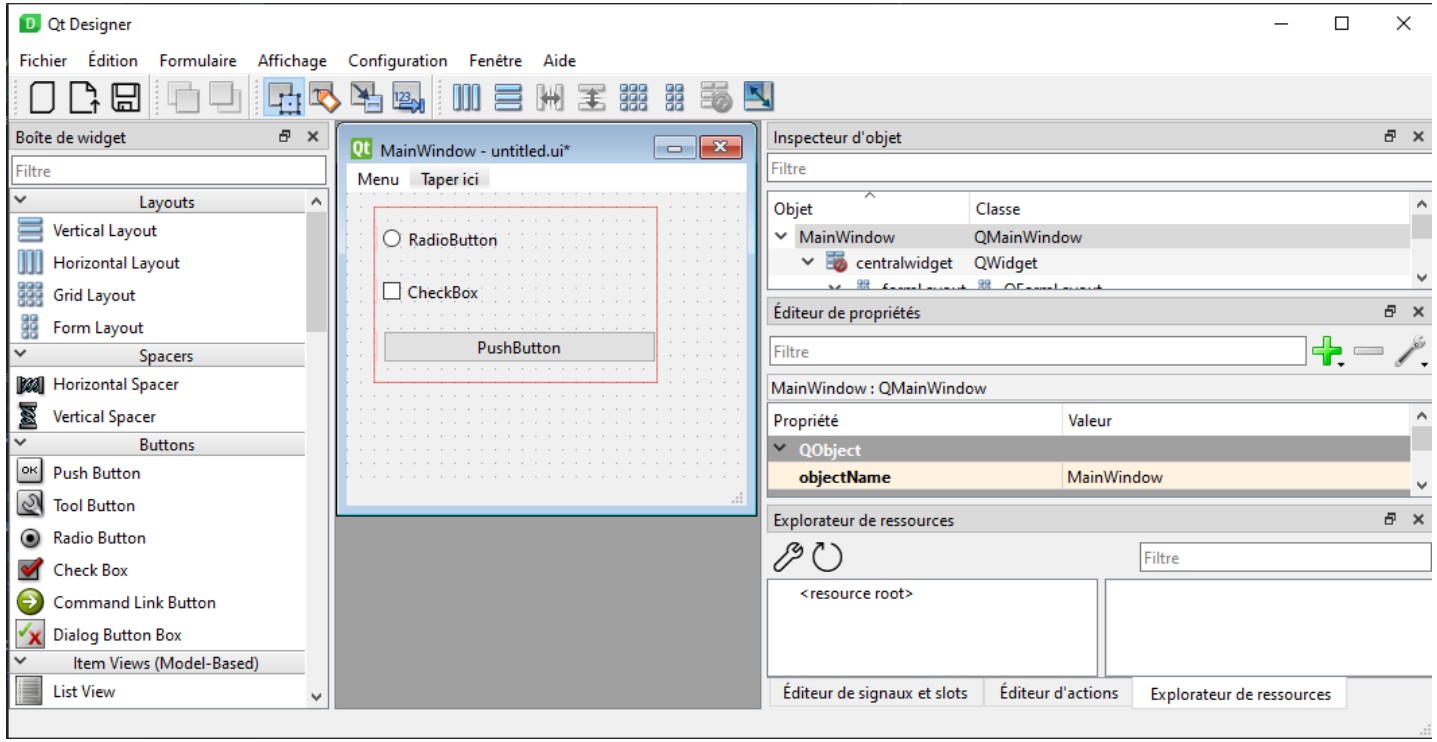


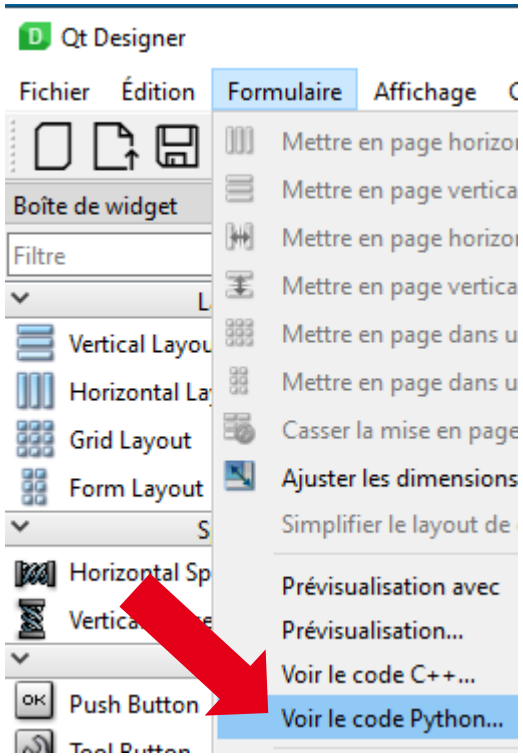
Initiation aux interfaces graphiques en Python

Utilisation QTDesigner (optionnel)

1. `pip install pyside6`

2. `python -c "import os, site, subprocess; subprocess.run([os.path.join(site.getsitepackages()[1], 'PySide6', 'designer')])"`





Initiation aux interfaces graphiques en Python

Exercices

Exercice 1 : Ajouter un champ Email

Objectif

Ajouter un champ de saisie d'email au formulaire.

Instructions

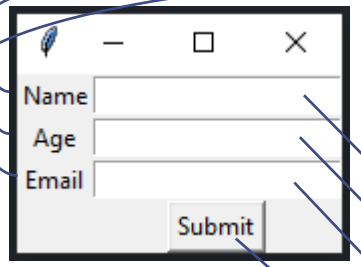
1. Ouvrez le fichier de formulaire pour le framework que vous utilisez :
   - Tkinter: form.py
   - PySide6: form.py
2. Ajoutez le champ email.
3. Mettez à jour la fonction `submit` pour afficher l'email dans la console.

Solutions

- Solution Tkinter
- Solution PySide6

Initiation aux interfaces graphiques en Python

Exercices




```python
import tkinter as tk

def submit():
    print(f"Name: {name_entry.get()}, Age: {age_entry.get()}, Email: {email_entry.get()}")

root = tk.Tk()
root.title("Simple Form")

tk.Label(root, text="Name").grid(row=0)
tk.Label(root, text="Age").grid(row=1)
tk.Label(root, text="Email").grid(row=2)

name_entry = tk.Entry(root)
age_entry = tk.Entry(root)
email_entry = tk.Entry(root)

name_entry.grid(row=0, column=1)
age_entry.grid(row=1, column=1)
email_entry.grid(row=2, column=1)

tk.Button(root, text="Submit", command=submit).grid(row=3, column=1)

root.mainloop()
```

Initiation aux interfaces graphiques en Python

Exercices

Exercice 2 : Ajouter une validation de formulaire

Objectif

Ajouter une validation aux champs du formulaire pour s'assurer que les champs nom, âge et email ne sont pas vides.

Instructions

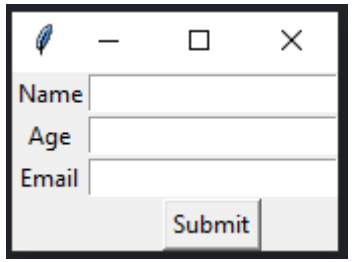
1. Ouvrez le fichier de solution pour l'Exercice 1 pour le framework que vous utilisez :
   - Tkinter : tkinter-solution.py
   - PySide6 : pyside6-solution.py
2. Ajoutez une validation pour s'assurer que les champs nom, âge et email ne sont pas vides.
3. Affichez un message d'erreur si la validation échoue.

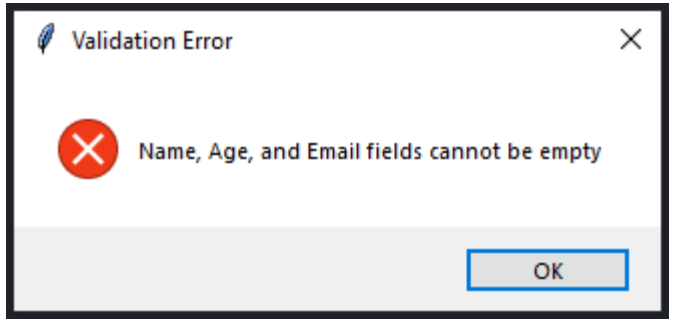
Solutions

- Solution Tkinter
- Solution PySide6

Initiation aux interfaces graphiques en Python

Exercices






```python
def submit():
    name = name_entry.get()
    age = age_entry.get()
    email = email_entry.get()

    if not name or not age or not email:
        messagebox.showerror("Validation Error", "Name, Age, and Email fields cannot be empty")
    else:
        print(f"Name: {name}, Age: {age}, Email: {email}")
```

Initiation aux interfaces graphiques en Python

Exercices

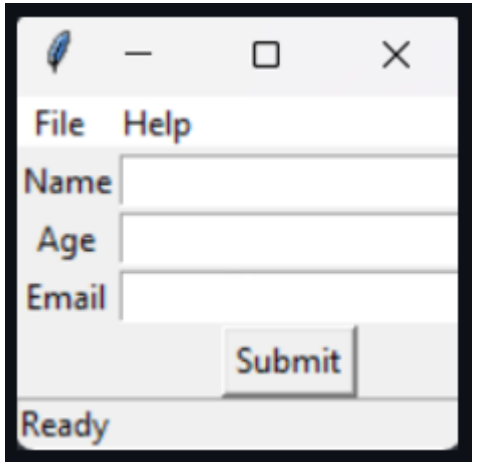
Exercice 3 : Ajouter un menu et une barre d'état

Objectif

Améliorer l'application en ajoutant un menu et une barre d'état.

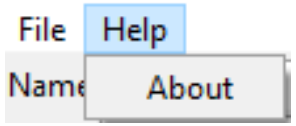
Instructions

1. Ouvrez le fichier corrigé de l'Exercice 2 correspondant au framework que vous utilisez :
   - Tkinter: tkinter-solution.py
   - PySide6: pyside6-solution.py
2. Ajoutez un menu avec les options suivantes :
   - `File` contenant une option `Quit` pour fermer l'application.
   - `Help` contenant une option `About` pour afficher un message d'information.
3. Ajoutez une barre d'état pour afficher un message par défaut.



Initiation aux interfaces graphiques en Python

Exercices

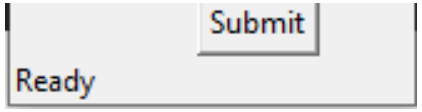


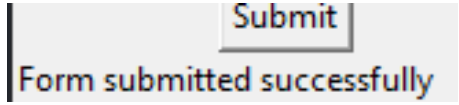

```python
menu_bar = tk.Menu(root)
root.config(menu=menu_bar)

file_menu = tk.Menu(menu_bar, tearoff=0)
menu_bar.add_cascade(label="File", menu=file_menu)
file_menu.add_command(label="Quit", command=quit_app)

help_menu = tk.Menu(menu_bar, tearoff=0)
menu_bar.add_cascade(label="Help", menu=help_menu)
help_menu.add_command(label="About", command=show_about)
```

```python
def submit():
```

⋮

```python
        print(f"Name: {name}, Age: {age}, Email: {email}")
        status_var.set("Form submitted successfully")

def quit_app():
    root.quit()

def show_about():
    messagebox.showinfo("About", "This is a simple form app
```






```python
status_var = tk.StringVar()
status_var.set("Ready")
status_bar = tk.Label(root, textvariable=status_var, bd=1, anchor=tk.W)
status_bar.grid(row=4, columnspan=2, sticky=tk.W+tk.E)
```

Sondage

bit.ly/3AMMMM

2. Introduction MVC

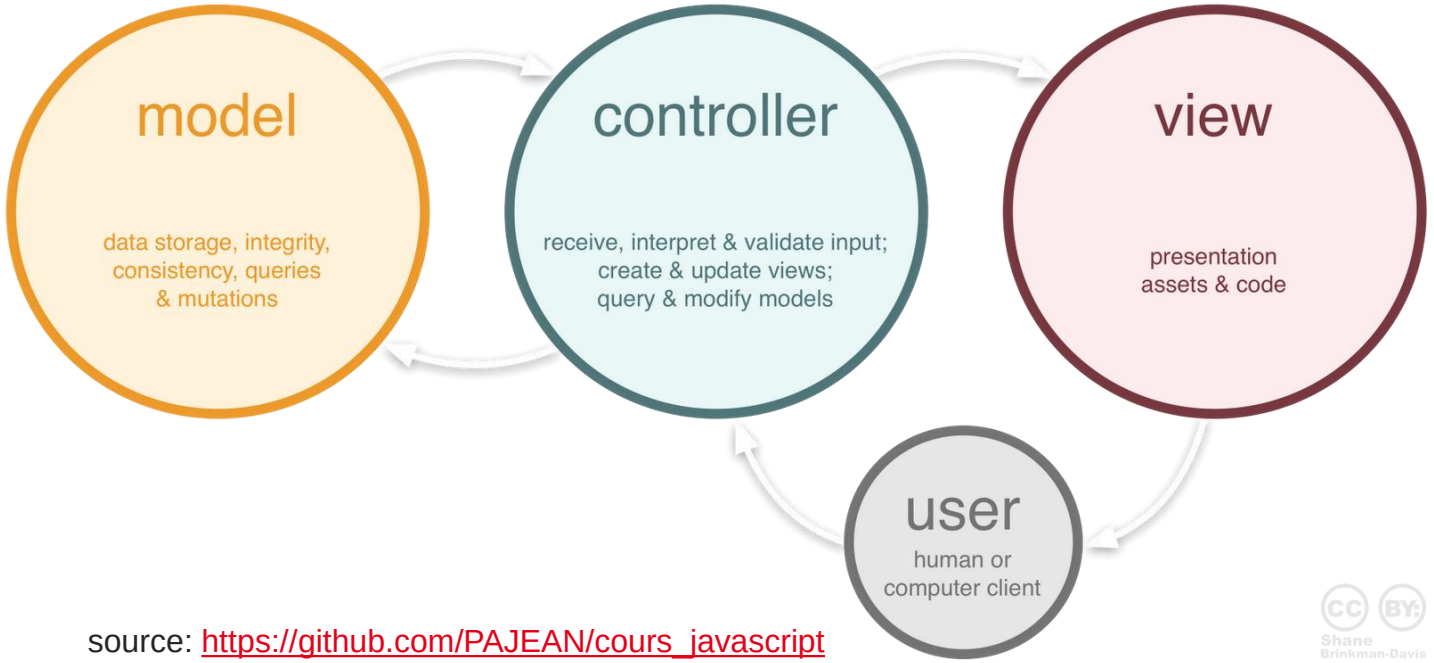
Introduction MVC

Explication du concept

Définition : Une méthode de structuration du code pour séparer la logique métier, l'affichage, et les interactions utilisateur.

Principe de base :

- **Modèle :** Gère les données et le contenu algorithmique
- **Vue :** Interface Graphique, permet à l'utilisateur d'intéragir avec le(s) modèle(s)
- **Contrôleur :** Relie l'utilisateur (la vue) et le système (le modèle).



source: https://github.com/PAJEAN/cours_javascript

Avantages :

- Code plus propre et maintenable.
- Facilite le travail en équipe.
- Simplifie les modifications futures.

Introduction MVC

Exemples

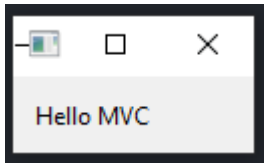
python-gui-mvc-tutorial / examples / mvc / simple / **helloworld.py**

PySide6

```python
class Model:
    def __init__(self):
        self.data = "Hello MVC"
```

```python
class View(QWidget):
    def __init__(self):
        super().__init__()
        self.setWindowTitle("Hello MVC App")

        self.layout = QVBoxLayout()
        self.label = QLabel("")
        self.layout.addWidget(self.label)
        self.setLayout(self.layout)

    def display(self, data):
        self.label.setText(data)
```

```python
class Controller:
    def __init__(self, model, view):
        self.model = model
        self.view = view
        self.update_view()

    def update_view(self):
        data = self.model.data
        self.view.display(data)
```



- class Model
  - func __init__
- class View
  - func __init__
  - func display
- class Controller
  - func __init__
  - func update_view

Introduction MVC

Exemples

python-gui-mvc-tutorial / examples / mvc / simple / count.py PySide6

```python
class Model:
    def __init__(self):
        self.count = 0

    def increment_count(self):
        self.count += 1
```

```python
class Controller:
    def __init__(self, model, view):
        self.model = model
        self.view = view
        self.view.button.clicked.connect(self.increment_count)
        self.update_view()

    def increment_count(self):
        self.model.increment_count()
        self.update_view()

    def update_view(self):
        count = self.model.count
        self.view.update_label(count)
```

```python
class View(QWidget):
    def __init__(self):
        super().__init__()
        self.setWindowTitle("Counter App")

        self.layout = QVBoxLayout()

        self.label = QLabel("Count: 0")
        self.layout.addWidget(self.label)

        self.button = QPushButton("Increment")
        self.layout.addWidget(self.button)

        self.setLayout(self.layout)

    def update_label(self, count):
        self.label.setText(f"Count: {count}")
```

- class Model
  - func __init__
  - func increment_count
- class View
  - func __init__
  - func update_label
- class Controller
  - func __init__
  - func increment_count
  - func update_view

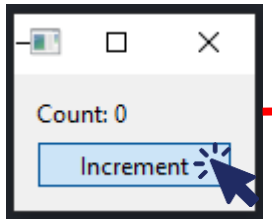
Introduction MVC

Exemples

python-gui-mvc-tutorial / examples / mvc / simple / **count.py** PySide6

class View

```
self.button = QPushButton("Increment")
```



1

class Controller

```
self.view.button.clicked.connect(self.increment_count)
```

2

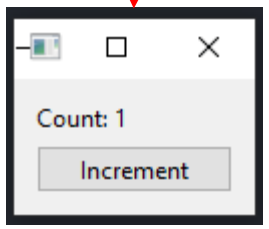
```
def increment_count(self):
    self.model.increment_count()
    self.update_view()

def update_view(self):
    count = self.model.count
    self.view.update_label(count)
```

3

class Model

```
def increment_count(self):
    self.count += 1
```

4

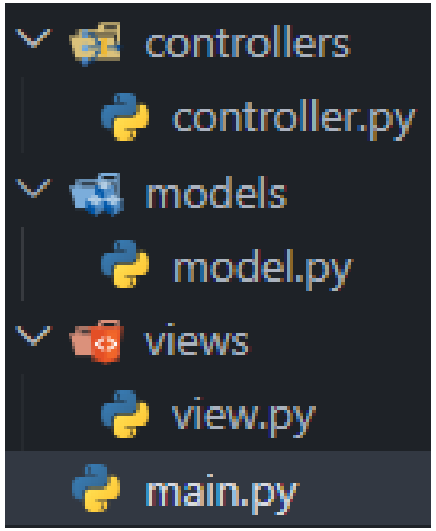


- class Model
  - func __init__
  - func increment_count
- class View
  - func __init__
  - func update_label
- class Controller
  - func __init__
  - func increment_count
  - func update_view

Introduction MVC

Exemples

python-gui-mvc-tutorial / examples / mvc / **full** /



main.py

```python
from PySide6.QtWidgets import QApplication
from models.model import Model
from views.view import View
from controllers.controller import Controller

if __name__ == "__main__":
    app = QApplication([])

    model = Model()
    view = View()
    controller = Controller(model, view)

    view.show()
    app.exec()
```

Liens Utiles

Lien repo GitHub: https://github.com/Crackvignoule/python-gui-mvc-tutorial

Documentation PySide6: https://doc.qt.io/qtforpython-6#documentation

Documentation Tkinter: https://docs.python.org/3/library/tkinter.html

PythonGUIs: Tkinter https://www.pythonguis.com/tkinter/

PythonGUIs: PySide6 https://www.pythonguis.com/pyside6/

PythonGUIs Examples https://github.com/pythonguis/pythonguis-examples

Which Python GUI Library Should You Choose? https://www.pythonguis.com/faq/which-python-gui-library/

superqt - A collection of custom widgets for PySide6 https://pyapp-kit.github.io/superqt/

PySide6 Examples https://doc.qt.io/qtforpython-6/examples/index.html

Merci

Des questions/remarques ?

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